

Advancing Scalable Particle-Based Reaction-Diffusion (PBRD) for the Whole-Cell Era.

Prof. Mark Flegg, Monash University

Whole-cell modelling (WCM) is a grand challenge for 21st-century science, demanding an interdisciplinary approach to create predictive tools that bridge the gap from fundamental molecular structures to the emergent behaviours of life. While a spatially- resolved model of a minimal cell has recently been proposed, its scalability to more complex cells remains a major hurdle. The most formidable computational bottleneck of these multi- physics, modular, WCMs lies in simulating stochastic reaction-diffusion processes. Reaction- Diffusion Master Equation (RDME) methods, for example the Spatial Gillespie Algorithm, are the most widely used approach but are fundamentally constrained by their reliance on discrete spatial domains. In contrast, Particle-Based Reaction-Diffusion (PBRD) simulations, while offering high spatial resolution and molecular-level stochasticity, have been historically overlooked in favour of RDME methods because the more significant scalability limitations of PBRD methods have typically been considered an insurmountable barrier to advancing WCMs. This talk will shed light on this scientific blind spot by introducing recent breakthroughs in PBRD methodologies that directly address and overcome its traditional scalability limitations. By exploring the mathematical foundations for how PBRD can efficiently simulate the molecular scale kinetics of increasingly complex cell types, this work pushes the boundaries of what is computationally feasible and paves the way for the next generation of whole-cell simulations.